Simplification of Mathematical Expressions

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What is a mathematical expression?

An expression in math is a mathematical statement that have a minimum of two terms containing numbers or variables, or both, connected by an operator in between.

Example: $2x + 7$ (Expression)

$9 + x = 10$ (Is an equation)

Types of Mathematical expressions

- Numerical expression

  This contains only numbers and mathematical operators.

  Eg: $10 - 5 + 2$

- Algebraic expression

What is an Algebraic Expression?

An algebraic expression in mathematics is an expression which is made up of variables and constants along with algebraic operations.

Eg: $5x^2 + x - 7$, $\frac{1}{3}xy - \frac{5}{7}$

Terms

Constant

$$5x \; - \; 3$$

Variable

Coefficient

Operator

Keywords

A variable is a symbol (often a letter) that is used to represent an unknown quantity.

Eg: x *or* y *or a etc.*

A coefficient is the value that is before a variable. It tells us how many lots of the variable there is.

Eg: $5x = x + x + x + x + x$

A term is a number by itself, a variable by itself, or a combination of numbers and letters.

Types of Algebraic expression

Monomial Expression

An algebraic expression which is having only one term is known as a monomial.

Eg: $3x^4, \ 3xy, \ 3x, \ 8y$

Binomial Expression

A binomial expression is an algebraic expression which is having two terms, which are unlike.

Eg: $x^3 + xyz, \ 5xy + 8$

Polynomial Expression

A polynomial expression consists of two or more algebraic terms.

Eg: $x^3 + 2x + 3$

Constructing algebraic expressions

Simple letters of the English alphabet such as $a, b, c, \dots, x, y, z$ are used to represent unknown constants and variables.

Example:

Let us denote the number of balls contain in a box in a shop by x. When two dozens of balls are sold, the number of balls remaining in the box can be denoted by $x - 24$.

The expression $x - 24$ is an **algebraic expression**.

Translating Words into Mathematical Expressions

Example:

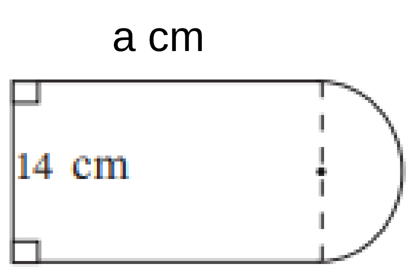
Phrase	Mathematical expression
Sum of y and 21	y+21
4 times q	$4q$
the speed u decreased by 20	$u - 20$
5 divided by a	$5/a$
Twice k	2k

Activity 1

Construct an algebraic expression for each of the following situations.

1. Martine is y years old. Jenny is 3 years younger than Martine. Connor is twice as older as Martine.
   a. Write an expression for Jenny’s age?
   b. Write an expression for Connor’s age?
   c. Write an expression for the sum of the three persons age.

2. The figure illustrates how a semi-circle has been joined to a rectangle of length a cm and breadth 14 cm such that the breadth of the rectangle is the diameter of the semi-circle. Find the perimeter of this figure in terms of a.



Solution

1. a) $y-3$ b) $2y$ c)$4y-3$
2. Since the arc length of a semi-circle of radius r is $\frac{1}{2}2\pi r$

The arc length of the semi-circle of radius 7 cm, $\frac{1}{2}\times 2\times\frac{22}{7}\times 7$

=22cm

The perimeter of the figure = a + a+ 14+ 22

=(2a + 36)cm

Methods of Simplifying expressions

1. Collecting like terms

   Eg: $5x + 3y + 4 - 2x + 8y - 7$

Steps:

Identify the like terms

The terms involving x are like terms. The terms involving y are like terms. The constant terms are like terms.

Group the like terms

$5x - 2x + 3y + 8y + 4 - 7$

Combine the like terms by adding or subtracting

$5x - 2x = 3x$

$3y + 8y = 11y$

$4 - 7 = -3$

So, $5x + 3y + 4 - 2x + 8y - 7 = 3x + 11y - 3$

Activity 2

Simplify each expression to lowest terms.

1. $5x + 8 - 2x + 5$

2. $3x + 5y - 9x + 7y$

3. $5xy + 6x^2 + 8xy - 9x^2$

4. $\sqrt{7}x + \sqrt{11}y + \sqrt{11} - \sqrt{5}y + x + 8$

Solution

1. $5x + 8 - 2x + 5$

$= 5x - 2x + 8 + 5$

$= 3x + 13$

2. 3x +5y -9x +7y

= 3x -9x +5y +7y

= -6x +12y

or

=12y -6x (Equivalent expressions)

3. $5x\,y + 6x^2 + 8x\,y - 9x^2$

$= 6x^2 - 9x^2 + 5x\,y + 8x\,y$

$= -3x^2 + 13x\,y$

4. $\sqrt{7}x + \sqrt{11}y + \sqrt{11} - \sqrt{5}y + x + 8$

$= \sqrt{7}x + x + \sqrt{11}y - \sqrt{5}y + \sqrt{11} + 8$

$= (\sqrt{7} + 1)x + (\sqrt{11} - \sqrt{5})y + \sqrt{11} + 8$

Methods of Simplifying expressions

2. Distributive Property

$$a(b + c) = ab + ac$$

Example:

Simplify: $5(2x + 5)$

$5(2x + 5) = 10x + 25$

Activity 3

Simplify each expression to lowest terms.

1. $5(3x + 4) - 7x + 2$

2. $2(2x - 3) + \sqrt{5}(9x - 5)$

3. $\frac{1}{3}(2x^2 + 3x - 3) - (4x^2 + 7x - 1)$

Solution

1. $5(3x+4)-7x+2$
$= 15x+20-7x+2$
$= 8x+22$

2. $2(2x-3)+\sqrt{5}(9x-5)$
$=4x-6+9\sqrt{5}x-5\sqrt{5}$
$=(4+9\sqrt{5})x-(6+5\sqrt{5})$

3. $\frac{1}{3}(2x^2+3x-3)-1(4x^2+7x-1)$
$=\frac{2}{3}x^2+\frac{1}{3}\times 3x-\frac{1}{3}\times 3-4x^2-7x+1$
$=\frac{2}{3}x^2+x-1-4x^2-7x+1$
$=\frac{2}{3}x^2-4x^2+x-7x-1+1$
$=-\frac{10}{3}x^2-6x$

Multiplication of Algebraic Expression

First lets look at a simple rule:

Product of powers with the same base

If x is a variable and m, n are positive integers, then

$$(x^m \times x^n) = x^{m+n}$$

Eg: $x^2.x = x.x.x = x^3$

1. Multiplication of two Monomials

Find the product of : $6xy$ and $-3x^2y^3$

$(6xy)(-3x^2y^3)$

$=\{6(-3)\} \text{ x } \{xy \text{ x } x^2y^3\}$

$= -18x^{1+2}\, y^{1+3}$

$= -18x^3y^4$

2. Multiplication of a polynomial by a Monomial

Find the product of $5a^2b^2$ and $(3a^2 - 4ab + 6b^2)$

$5a^2b^2$ x $(3a^2 - 4ab + 6b^2)$

$= 5a^2b^2 \times 3a^2 + 5a^2b^2 \times (-4ab) + 5a^2b^2 \times (6b^2)$

$= 15a^{(2+2)}b^2 - 20a^{(2+1)}b^{(2+1)} + 30a^2b^{(2+2)}$

$= 15a^4b^2 - 20a^3b^3 + 30a^2b^4$

3. The product of two binomial expressions

Multiply the terms within the second pair of brackets by the two terms within the first pair of brackets.

$$(x+3)(x+2) = (x+3)(x+2)$$

$$= x(x+2) + 3(x+2)$$

$$= x^2 + 2x + 3x + 6$$

$$= x^2 + 5x + 6$$

4. Multiplication by Polynomials

Multiply $(5x^2 - 6x + 9)$ by $(2\ x\ -3)$

$(2\ x\ -3)\ (5x^2 - 6x + 9)$

$= 2\ x\ (5x^2 - 6x + 9)\ -3(5x^2 - 6x + 9)$

$= 2\ x\ (5x^2) + 2\ x(-6x) + 2\ x(9) - 3(5x^2) - 3(-6x) - 3(9)$

$= 10x^3 - 12x^2 + 18\ x - 15x^2 + 18\ x - 27$

$= 10x^3 - 12x^2 - 15x^2 + 18\ x + 18\ x - 27$

$= 10x^3 - 27x^2 + 36\ x - 27$

Activity 4

Simplify each expression to lowest terms.

1. $(x+y)(x-y)$

2. $(5x\ -4)^2$

3. $(x-y)^3$

4. $(x+y)^4$

Solution

1 . $(x + y)(x - y) = x\ (x - y) + y\ (x - y\ )$

$$= x^2 - xy + yx - y^2$$

$$= x^2 - y^2$$

2. $(5x - 4)^2 = (5\ x - 4)\ (5\ x - 4)$

$$= 5\ x(5\ x - 4)\ -4(5\ x - 4)$$

$$= 25x^2 - 20x - 20x + 16$$

$$= 25x^2 - 40x + 16$$

Solution

3. $(x-y)^3 = (x-y)(x-y)(x-y)$

$$= (x-y)(x^2-2xy+y^2)$$

$$= x(x^2-2xy+y^2) - y(x^2-2xy+y^2)$$

$$= x^3-2x^2y+xy^2-yx^2+2xy^2-y^3$$

$$= x^3-3x^2y+3xy^2-y^3$$

4. $(x+y)^4 = (x+y)^2(x+y)^2$

$$= x^4+4x^3y+6x^2y^2+4xy^3+y^4$$

Division of Algebraic Expression

First let us look at quotient law,

In division of algebraic expression if x is a variable and m, n are positive integers such that $m > n$ then $\frac{x^m}{x^n} = x^{(m-n)}$.

Eg: $\frac{x^5}{x^3} = \frac{x.x.\cancel{x.x.x.}}{\cancel{x.x.x}} = x^2$

1. Division of a Monomial by a Monomial

Example: $\frac{12x^3y^2}{3x^2y} = \frac{12}{3}\,\frac{x^3}{x^2}\,\frac{y^2}{y} = 4x^{(3-2)}y^{(2-1)} = 4xy$

$$= \frac{12\,.x\,.\,x\,.\,x\,.y\,.y}{3\,.x\,.x\,\,.y} = 4xy$$

2. Division of a Polynomial by a Monomial

We are dividing each term of the given polynomials by the monomials.

Example: Divide $x^3y^3 + x^2y^3 - xy^4 + xy$ by xy

$$\frac{x^3y^3 + x^2y^3 - xy^4 + xy}{xy}$$

$$= \frac{x^3y^3}{xy} + \frac{x^2y^3}{xy} - \frac{xy^4}{xy} + \frac{xy}{xy}$$

$$= x^2y^2 + xy^2 - y^3 + 1$$

3. Division of a Polynomial by a Polynomial

Example: Divide $y^2 + 18y + 65$ by $y + 5$

Solution: $\frac{y^2+18y+65}{y+5}$;

$$\frac{y^2+18y+65}{y+5} = (y + 13)$$

Y +13 ← Quotient

y+5 $\big)$ $y^2 + 18y + 65$ ← Dividend

Divisor ↑

$y^2 +$ 5y

13y +65

13y+ 65

0

↑

Remainder

Activity 5

Simplify each expression to lowest terms.

1. $\frac{28x^5y^4}{7x^8y^{-3}}$

2. $(2x^2 - 7x + 4)(3x^2 + 5x + 7)$

Solution

1. $\frac{28x^5y^4}{7x^8y^{-3}} = 4\ x^{-3}\ y^7 = \ 4\frac{y^7}{x^3}$

2. $(2x^2\ - 7x + 4)(3x^2 + 5x + 7)$

$= 2x^2\ (3x^2 + 5x + 7)\ -7x(3x^2 + 5x + 7)+4\ (3x^2 + 5x + 7)$

$=6x^4 + 10\ x^3 +14x^2\ - 21x^3\ - 35x^2 - 49x +\ 12x^2 + 20x + 28$

$=6x^4 + 10\ x^3 -21x^3 + 14x^2 -35x^2 + 12x^2 -49x + 20x + 28$

$=6x^4$ $-11x^3\ - 9x^2 - 29x + 28$